

MATRICES

Addition $\begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} a+e & b+f \\ c+g & d+h \end{bmatrix}$

* ต้องเป็น matrix ขนาดเดียวกัน only!

Multiplication SCALAR $4 \cdot \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 4a & 4b \\ 4c & 4d \end{bmatrix}$

MATRICES $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \times \begin{bmatrix} g & h \\ i & j \end{bmatrix} = \begin{bmatrix} ag+bi & ah+bj \\ cg+di & ch+dj \end{bmatrix}$ * column 1 = row 2 only!
(ถ้าไม่เท่ากัน = undefined)

Matrix representation: $\begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix}$ Column $m \times n$

PROPERTIES

$A+B = B+A$
 $A+(B+C) = (A+B)+C$
 $(cA)A = (cA)A$
 $1A = A$
 $(A+B)C = (A+B)C$
 $(C+D)A = C(A+D)$
 $(AB)C = A(BC)$
 $A(B+C) = AB+AC$
 $C(AB) = (CA)B = A(CB)$

IDENTITY MATRIX

Identity matrix $I_n = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ 0 & \dots & 1 & 0 \\ 0 & \dots & 0 & 1 \end{bmatrix}$

$AI_n = A$
 $I_m A = A$ (เมื่อ A มีขนาด $m \times n$)

TRANSPOSE

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow A' = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$
 $B = \begin{bmatrix} g & h \\ i & j \end{bmatrix} \rightarrow B' = \begin{bmatrix} g & i \\ h & j \end{bmatrix}$

PROPERTIES

$(A')^T = A$
 $(A+B)^T = A^T + B^T$
 $(cA)^T = c(A^T)$
 $(AB)^T = B^T A^T$ * ต้องจำลำดับที่กลับ

INVERSE MATRIX

$AB = BA = I_n \rightarrow B$ เป็น inverse ของ A
 $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow A^{-1} = \frac{1}{\det A} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$
 For matrix $n \times n$:
 $A^{-1} = \frac{1}{\det A} (adj A)$
 $adj(A) = [C_{ij}]^T$
 $C_{ij} = (-1)^{i+j} M_{ij}$ (for matrix 2×2 only!)

PROPERTIES

$(A^{-1})^{-1} = A$ | $(A^T)^{-1} = (A^{-1})^T$
 $(A^k)^{-1} = A^{-1} \cdot A^{-1} \cdot \dots \cdot A^{-1} = (A^{-1})^k$
 $(cA)^{-1} = \frac{1}{c} (A^{-1})$; $c \neq 0$ | $(AB)^{-1} = B^{-1} A^{-1}$

DETERMINANT *square matrix only!

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow \det A = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$
 $B = \begin{bmatrix} a & b & g \\ c & d & h \\ e & f & i \end{bmatrix} = \begin{vmatrix} a & b & g \\ c & d & h \\ e & f & i \end{vmatrix} = (ad + bhe + gcf) - (edg + fha + icb)$

$\det A = |A| = \sum_{j=1}^n a_{1j} C_{1j} = a_{11} C_{11} + a_{12} C_{12} + \dots + a_{1n} C_{1n}$

CRYPTOGRAPHY ถอดรหัสข้อความ

0 = —	7 = G	14 = N	21 = U
1 = A	8 = H	15 = O	22 = V
2 = B	9 = I	16 = P	23 = W
3 = C	10 = J	17 = Q	24 = X
4 = D	11 = K	18 = R	25 = Y
5 = E	12 = L	19 = S	26 = Z
6 = F	13 = M	20 = T	

PROPERTIES

$\det(A) \neq 0$
 $\det(AB) = \det(A) \det(B)$
 $\det(cA) = c^n \det(A)$ n = ขนาดเมทริกซ์
 $\det(A^{-1}) = \frac{1}{\det(A)}$ | $\det(A) = \det(A^T)$

ถ้าเจอ matrix ที่ไม่มีอินเวอร์สแทนที่ $\det = 0$

$\begin{bmatrix} 0 & 0 & 0 \\ a & b & c \\ d & e & f \end{bmatrix} \rightarrow \begin{bmatrix} a & b & 3a \\ d & e & 3d \\ g & h & 3g \end{bmatrix}$

MEET - ME - M O N D A Y -

13 5 5 | 20 0 13 | 5 0 13 | 15 14 4 | 25 0

encode : [message] $\times$ [matrix]

decode : [message] $\times$ [inverse matrix]

CARMER'S RULE หากทำ X, Y, Z

ex. (2 ตัวแปร)

$\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases} \Rightarrow A = \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix}, X = \begin{bmatrix} x \\ y \end{bmatrix}, D = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$
 $x = \frac{\det \begin{bmatrix} c_1 & b_1 \\ c_2 & b_2 \end{bmatrix}}{\det A}, y = \frac{\det \begin{bmatrix} a_1 & c_1 \\ a_2 & c_2 \end{bmatrix}}{\det A}$

ex. (3 ตัวแปร)

$\begin{cases} a_1x + b_1y + c_1z = d_1 \\ a_2x + b_2y + c_2z = d_2 \\ a_3x + b_3y + c_3z = d_3 \end{cases} \Rightarrow A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, D = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$
 $x = \frac{\det A}{\det A}, y = \frac{\det A}{\det A}, z = \frac{\det A}{\det A}$

GAUSSIAN ELIMINATION

$\begin{cases} a_1x + b_1y + c_1z = d_1 \\ a_2x + b_2y + c_2z = d_2 \\ a_3x + b_3y + c_3z = d_3 \end{cases} \Rightarrow A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$ - ต้องทำจนกว่าเป็น matrix ขั้นบันได $\rightarrow \begin{bmatrix} 1 & \dots & \dots \\ 0 & 1 & \dots \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{cases} x + y + z = \dots \\ y + z = \dots \\ z = \dots \end{cases}$ แล้วแก้หา x, y, z

GAUSSIAN-JORDAN

$\begin{cases} a_1x + b_1y + c_1z = d_1 \\ a_2x + b_2y + c_2z = d_2 \\ a_3x + b_3y + c_3z = d_3 \end{cases} \Rightarrow A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$ - ต้องทำจนกว่าเป็น identity matrix $\rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{cases} x = \dots \\ y = \dots \\ z = \dots \end{cases}$